Final check of the evolution terms: three and four ρ's, two ρ's, JIMWLK, BFKL and DGLAP

Files checked: `Evolution_terms.tex`, `Evolution_three_rho.tex`, `Group_I.tex`, `Group_II.tex`, `Group_III.tex` (this upload), and `4rho.tex` (earlier upload; not sent again). Notation as in your files: $\mathcal{K}(X) = X/X^2$, $K(a,b,z) = \mathcal{K}(a-z)\cdot\mathcal{K}(b-z)$, $U^{bc} = U^{cb}$, and

$$\mathcal{N} = \frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+} = \frac{\alpha_s^2}{8\pi^6 k^+}, \qquad \text{so that} \qquad \mathcal{N}\log\frac{\vee}{\Lambda} \;\leftrightarrow\; \frac{\alpha_s^2\,\delta Y}{8\pi^6(N_c^2-1)} \text{ of your JIMWLK and BFKL formulas}$$

(with $\langle\rho^a\rho^b\rangle = \delta^{ab}\langle\rho^f\rho^f\rangle/(N_c^2-1)$ and $\int_{w,w'} e^{-ik\cdot(w'-w)}$). In this note blue is new or corrected text and red is removed text. The two-ρ terms are coloured by their kernel:

- **green, JIMWLK only:** the unobserved gluon z is attached to the measured gluon (w or w'). These kernels occur only in JIMWLK⊗LO.
- **purple, charge–charge:** the unobserved gluon z is exchanged between the two charges (x, x'). These kernels occur in both JIMWLK⊗LO and BFKL⊗LO, with different colour structures.
- **orange, BFKL only:** the measured gluon is emitted by the gluon at z. These kernels occur only in BFKL⊗LO.

1 Summary

Results.

1. **Four ρ's cancel.** After symmetrization the four-ρ terms $\propto \log\frac{\vee}{\Lambda}$ cancel exactly (`4rho.tex`; residual 4.7×10^{-15} of single terms).
2. **Three ρ's cancel.** The list "All three ρ term with c.c." of `Evolution_three_rho.tex` (152 terms) is identical to the list checked before. After symmetrization it cancels exactly: 0 non-cancelling kernel monomials, residual 2×10^{-15} (Sec. 7).
3. **Two ρ's $\propto \log\frac{\vee}{\Lambda}$.**
   - Sec. 2 of `Evolution_terms.tex` (the 335 terms from reordering three and four ρ's) is correct. Row by row it agrees with the checked lists: 0 differing kernel monomials in every row, pointwise 6×10^{-15}.
   - Sec. 1 (genuine terms) is correct (fixes F5, F6 are in). In Row V 2b, the UV term and the finite counterterm cancel each other exactly (Sec. 4.1).
   - **Missing (mistake M1):** the two-ρ terms of `4rho.tex` ("Two ρ contribution", with fixes F1, F2) are not in `Evolution_terms.tex`. Without them, 6 of the 26 kernel monomials differ from JIMWLK⊗LO (rel. 1.7×10^{-1}). The differing structures are $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$, $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$, $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ (18 of 152 independent colour structures). They are written out in Eq. (4.9).
   - **With them, the sum is JIMWLK⊗LO exactly.** This holds in all 13 kernel structures and all 154 independent colour structures, with exact N_c coefficients (0 differences); pointwise 6×10^{-16}.
4. **Your JIMWLK formula (mistake M2).** As sent it has two errors: the matrix M sits at the wrong points in $dN^{(1)}$–$dN^{(4)}$, and $dN^{(5)}$ has the wrong sign. As sent it differs from JIMWLK⊗LO by 0.22–0.47 (relative); corrected (Eq. (3.1)) it agrees to 2×10^{-15}.
5. **BFKL.** There is no BFKL part in the $\log\frac{\vee}{\Lambda}$ terms. BFKL⊗LO is the coefficient of $\log\vee$, i.e. the sum of the coefficients of $\log\frac{\vee}{\Lambda}$ and of $\log\frac{\vee}{k^+}$ (Sec. 2). That sum is BFKL⊗LO exactly: 9 kernel structures, 109 independent colour structures, 0 differences (Sec. 5). Your BFKL formula (App. E) is correct as sent (8×10^{-15}).
6. **DGLAP.** The terms with poles are the collinear ones (Sec. 6).
   - In the $\log\frac{\vee}{\Lambda}$ sector there are three (Group I Row II, Group I Row V 2b UV, Group II Row IV). Their poles, γ_E and μ cancel exactly; a finite logarithm is left, which belongs to JIMWLK⊗LO.
   - The DGLAP evolution of the fragmentation function comes from Group II Row IV at general ξ. With Group I and the charge renormalization it gives the full P_{gg}.

Mistakes in the cross-section files (not in the evolution terms).

M3 `Group_I.tex`, Eq. (1.5) (Row II part 2a). The $\mathfrak{F}_\beta$ change is now in, but:

- in the second term of the first bracket, $(w-z)\cdot(z-x)$ must be $(w-z)\cdot(x-z)$. As written the bracket is symmetric under $x\leftrightarrow y$, so it vanishes against $f^{hbc}\{\rho^b(x),\rho^c(y)\}$;
- the third term (the one with $f^{abc}U^{bd}(x)U^{ce}(y)$) must have the overall sign $-$, not $+$: with $\mathfrak{F}=-\pi[\cdots]$, $-\frac{ig^4}{32\pi^5}\big[\log\frac{\vee}{\Lambda}-\log\frac{\vee}{k^+}\big]\mathfrak{F}\times(-f^{abc}UU)=-\frac{ig^4}{32\pi^4}\big[\log\frac{\vee}{\Lambda}-\log\frac{\vee}{k^+}\big][\cdots]f^{abc}UU$.

M4 `Evolution_three_rho.tex`, the row-by-row display at the top (not the list that is used):

- Row II still has the full $\mathfrak{F}$ in the $U^{ah}(w)f^{hbc}$ term (should be $\mathfrak{F}_\beta$, F4);
- Row III, second part, is still in its original form; the list uses $\mathcal{D}$ (F3).

The list "All three ρ term with c.c." has both corrections and cancels.

Final answer.

$$\boxed{\frac{d^3N_{\rm LL}}{d^3k}=\mathcal{N}\log\frac{\vee}{\Lambda}\,\mathcal{J}+\mathcal{N}\log\frac{\vee}{k^+}\,(\mathcal{B}-\mathcal{J})=\mathcal{N}\Big[\log\frac{k^+}{\Lambda}\,\mathcal{J}+\log\frac{\vee}{k^+}\,\mathcal{B}\Big]} \tag{1.1}$$

- $\mathcal{J}$ is JIMWLK⊗LO: Eq. (3.1) (= Eq. (4.25)). It is the sum of all two-ρ terms $\propto\log\frac{\vee}{\Lambda}$ (Sec. 4).
- $\mathcal{B}$ is BFKL⊗LO: Eq. (3.2) (= Eq. (5.15)). It is the sum of the coefficients of $\log\vee$ (Sec. 5).
- The target (JIMWLK) evolves over $\log\frac{k^+}{\Lambda}$ and the projectile (BFKL) over $\log\frac{\vee}{k^+}$.
- DGLAP for the fragmentation function: Eq. (6.4).

2 Why $\log\frac{\vee}{\Lambda}$ carries JIMWLK and not BFKL

Every row is an exact p^+ integral of the form $\int dp^+\,F(p^+)$.

- Λ enters only through the limit where the unobserved gluon is softer than the measured one ($p^+\to\Lambda$). There $F\to\mathfrak{T}/p^+$.
- $\vee$ enters only through the upper limit, where the unobserved gluon is harder than the measured one ($p^+\to\vee$). There $F\to\mathfrak{P}/p^+$.

So, exactly and with unique coefficients,

$$\int dp^+F(p^+)=\mathfrak{T}\,\log\frac{k^+}{\Lambda}+\mathfrak{P}\,\log\frac{\vee}{k^+}+R=\mathfrak{T}\,\log\frac{\vee}{\Lambda}+(\mathfrak{P}-\mathfrak{T})\log\frac{\vee}{k^+}+R, \tag{2.1}$$

where R contains neither logarithm. With your convention $\log\frac{\Lambda}{k^+}=-\log\frac{\vee}{\Lambda}+\log\frac{\vee}{k^+}$:

- the coefficient of $\log\frac{\vee}{\Lambda}$ is the soft end $\mathfrak{T}$ (the gluon is softer than the measured one). A soft gluon evolves the target: summed over all rows, it is JIMWLK⊗LO;
- the hard end $\mathfrak{P}$ (the measured gluon is soft compared with the unobserved one) evolves the projectile. It is BFKL⊗LO, and it multiplies $\log\vee$: the $\log\frac{\vee}{\Lambda}$ and $\log\frac{\vee}{k^+}$ terms together.

This gives Eq. (1.1). The colours show it term by term. Every orange kernel (the measured gluon emitted by the gluon at z) is absent from the $\log\frac{\vee}{\Lambda}$ terms, and every green kernel is absent from the $\log\vee$ terms. Equivalently, $\frac{d^3N_{\rm LL}}{d^3k}=\mathcal{N}\big[\frac12\log\frac{\vee}{\Lambda}(\mathcal{J}+\mathcal{B})+\frac12\log\frac{k^{+2}}{\vee\Lambda}(\mathcal{J}-\mathcal{B})\big]$: $\log\frac{\vee}{\Lambda}$ multiplies $\frac12(\mathcal{J}+\mathcal{B})$ only if $k^+=\sqrt{\vee\Lambda}$.

3 The targets

3.1 JIMWLK⊗LO: your formula, corrected (M2)

With $P\equiv\frac{\alpha_s^2\delta Y}{8\pi^6(N_c^2-1)}$ and $\int\equiv\int_{x,y,t}\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\langle\rho^f(x)\rho^f(y)\rangle$, the rule is: in a term $\mathcal{K}(u,v,t)\,{\rm Tr}\{T^eU^\dagger(v)U(u)T^e\,M\}$,

$$M=(U^\dagger(u)-U^\dagger(t))(U(v)-U(t)),$$

i.e. M sits at the two points the colour charges act on. As sent, M was always $(U^\dagger(x)-U^\dagger(t))(U(y)-U(t))$ in $dN^{(1)}$, $dN^{(3)}$ and $(U^\dagger(y)-U^\dagger(t))(U(x)-U(t))$ in $dN^{(2)}$, $dN^{(4)}$. The corrected formula:

$$\frac{dN^{(1)}}{d^2w d^2w' dy}=P\int\Big[$$

$$
\begin{aligned}
&\mathcal{K}(y,x,t)\mathrm{Tr}\{T^e U^\dagger(x)U(y)T^e(U^\dagger(y)-U^\dagger(t))(U(x)-U(t))\}\\
&-\mathcal{K}(w',x,t)\mathrm{Tr}\{T^e U^\dagger(x)U(w')T^e(U^\dagger(w')-U^\dagger(t))(U(x)-U(t))\}\\
&-\mathcal{K}(y,w,t)\mathrm{Tr}\{T^e U^\dagger(w)U(y)T^e(U^\dagger(y)-U^\dagger(t))(U(w)-U(t))\}\\
&+\mathcal{K}(w',w,t)\mathrm{Tr}\{T^e U^\dagger(w)U(w')T^e(U^\dagger(w')-U^\dagger(t))(U(w)-U(t))\}\Big],
\end{aligned}
$$

$$
\begin{aligned}
\frac{dN^{(2)}}{d^2w d^2w' dy} = P\int\Big[\;&\\
&\mathcal{K}(x,y,t)\mathrm{Tr}\{T^e U^\dagger(x)U(y)T^e(U^\dagger(y)-U^\dagger(t))(U(x)-U(t))\}\\
&-\mathcal{K}(x,w',t)\mathrm{Tr}\{T^e U^\dagger(x)U(w')T^e(U^\dagger(w')-U^\dagger(t))(U(x)-U(t))\}\\
&-\mathcal{K}(w,y,t)\mathrm{Tr}\{T^e U^\dagger(w)U(y)T^e(U^\dagger(y)-U^\dagger(t))(U(w)-U(t))\}\\
&+\mathcal{K}(w,w',t)\mathrm{Tr}\{T^e U^\dagger(w)U(w')T^e(U^\dagger(w')-U^\dagger(t))(U(w)-U(t))\}\Big],
\end{aligned}
$$

$$
\begin{aligned}
\frac{dN^{(3)}}{d^2w d^2w' dy} = -P\int\Big[&\mathcal{K}(y,y,t)\mathrm{Tr}\{T^e(U^\dagger(x)-U^\dagger(w))U(y)T^e(U^\dagger(y)-U^\dagger(t))(U(y)-U(t))\}\\
&-\mathcal{K}(w',w',t)\mathrm{Tr}\{T^e(U^\dagger(x)-U^\dagger(w))U(w')T^e(U^\dagger(w')-U^\dagger(t))(U(w')-U(t))\}\Big],
\end{aligned}
$$

$$
\begin{aligned}
\frac{dN^{(4)}}{d^2w d^2w' dy} = -P\int\Big[&\mathcal{K}(x,x,t)\mathrm{Tr}\{T^e U^\dagger(x)(U(y)-U(w'))T^e(U^\dagger(x)-U^\dagger(t))(U(x)-U(t))\}\\
&-\mathcal{K}(w,w,t)\mathrm{Tr}\{T^e U^\dagger(w)(U(y)-U(w'))T^e(U^\dagger(w)-U^\dagger(t))(U(w)-U(t))\}\Big],
\end{aligned}
$$

$$
\begin{aligned}
\frac{dN^{(5)}}{d^2w d^2w' dy} = -+P\int\Big[\;&\\
&\mathcal{K}(y,y,t)\mathrm{Tr}\{(U^\dagger(y)-U^\dagger(t))U(y)T^d\}\mathrm{Tr}\{(U^\dagger(x)-U^\dagger(w))U(y)T^d\}\\
&-\mathcal{K}(w',w',t)\mathrm{Tr}\{(U^\dagger(w')-U^\dagger(t))U(w')T^d\}\mathrm{Tr}\{(U^\dagger(x)-U^\dagger(w))U(w')T^d\}\\
&-\mathcal{K}(x,x,t)\mathrm{Tr}\{(U^\dagger(x)-U^\dagger(t))U(x)T^d\}\mathrm{Tr}\{U^\dagger(x)(U(y)-U(w'))T^d\}\\
&+\mathcal{K}(w,w,t)\mathrm{Tr}\{(U^\dagger(w)-U^\dagger(t))U(w)T^d\}\mathrm{Tr}\{U^\dagger(w)(U(y)-U(w'))T^d\}\Big],
\end{aligned}
\tag{3.1}
$$

and $\mathcal{J}=\sum_{i=1}^{5} dN^{(i)}$ (per unit δY). After the correction, $dN^{(2)}$ is identical to $dN^{(1)}$ term by term. Numerically, at random points with random adjoint Wilson lines, $\sum_i dN^{(i)}$ equals JIMWLK⊗LO to 2×10^{-15}; as sent it differs by 0.22–0.47 (relative).

3.2 BFKL⊗LO: your formula (correct as sent)

$$
\begin{aligned}
\frac{dN}{d^2w\,d^2w'\,dy}\Big|_{\mathrm{BFKL}} = -\frac{\alpha_s^2 N_c\,\delta Y}{8\pi^6(N_c^2-1)}\int_{x,y,z}\langle\rho^f(x)\rho^f(y)\rangle\Big[\;&\\
&\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')K_D(x,y,z)\mathrm{Tr}\{(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\}\\
&-2\,\mathcal{K}(z-w)\cdot\mathcal{K}(z-w')K(x,y,z)\mathrm{Tr}\{(U^\dagger(z)-U^\dagger(w))(U(z)-U(w'))\}\\
&+\mathcal{K}(z-w)\cdot\mathcal{K}(y-w')\big(2K(x,y,z)-K(x,x,z)\big)\mathrm{Tr}\{(U^\dagger(z)-U^\dagger(w))(U(y)-U(w'))\}\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(z-w')\big(2K(x,y,z)-K(y,y,z)\big)\mathrm{Tr}\{(U^\dagger(x)-U^\dagger(w))(U(z)-U(w'))\}\Big],
\end{aligned}
\tag{3.2}
$$

with $K_D(x,y,z)=K(x,x,z)-2K(x,y,z)+K(y,y,z)$. It agrees with BFKL⊗LO to 8×10^{-15}.

4 All two-ρ terms proportional to $\log\frac{\vee}{\Lambda}$

Number of terms (green / purple): genuine, expanded: 135 / 4; `4rho.tex` lists: 0 / 30; your Sec. 2: 208 / 127. No orange term appears.

4.1 Genuine terms (Sec. 1 of `Evolution_terms.tex`)

To compare term by term, every dimensionally regulated bracket is written back as the z integral it comes from (exact in $d_\perp=2-\epsilon$, $\overline{\mathrm{MS}}$):

- Group I Row II: $\frac{2}{\epsilon}+2\gamma+\log\frac{(x-w)^2\mu^2}{4}=-\frac{1}{\pi}\int_z\big[K(x,w,z)-K(w,w,z)\big]$;
- Group I Row V 2b UV: $\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y-z)^2\mu^2}{4}=\frac{1}{\pi}\int_x\big[K(z,z,x)-K(y,z,x)\big]$;
- Group II Row IV: $-\frac{d_\perp}{\epsilon}-d_\perp\gamma_E-\frac{d_\perp}{2}\log\frac{(y'-y)^2\mu^2}{4}-1=\frac{1}{\pi}\int_z\big[K(y,y',z)-\frac{1}{2}K(y,y,z)-\frac{1}{2}K(y',y',z)\big]+O(\epsilon)$.

The terms are expanded, the charges are named x (emission ending at w) and x' (ending at w'), and the colour algebra is done. Group I and Group III include their c.c.

- **Row V 2b: the UV term and the finite counterterm cancel each other.** Both have the kernel $\mathcal{K}(x'-w')\cdot\mathcal{K}(y-z)\,[K(z,z,x)-K(y,z,x)]$ and the colour $N_c[U(y)-U(z)]$.
  - UV: $+\frac{g^4N_c}{8\pi^4}\cdot\frac{1}{\pi}\int_x$.
  - Counterterm: $-\frac{g^4}{8\pi^5}N_c$, pointwise in x.

  So the $\log\frac{\vee}{\Lambda}$ part of Row V 2b is just its two endpoint terms. This is the same as before the UV subtraction.
- Your genuine terms are equal to the corrected ones of the earlier check: Group III Row III with the sign F5, Group I Row II with the bracket of F6 (numerically, rel. $\leq 10^{-15}$).

Group I, Row II.

$$
\begin{aligned}
\left.\frac{d^3N}{d^3k}\right|_{G1\text{-}II} = \mathcal{N}\log\frac{\vee}{\Lambda}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[& \\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\Big]
\end{aligned}
\tag{4.1}
$$

Group I, Row V, part 2a.

$$
\begin{aligned}
\left.\frac{d^3N}{d^3k}\right|_{G1\text{-}V\ 2a} = \mathcal{N}\log\frac{\vee}{\Lambda}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[& \\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{be}(z)\,U^{ch}(w)\,\rho^l(x')\,\rho^d(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{be}(z)\,U^{ch}(w)\,\rho^l(x')\,\rho^d(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(x')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(w')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{be}(z)\,U^{ch}(w')\,\rho^d(x')\,\rho^l(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{be}(z)\,U^{ch}(w')\,\rho^d(x')\,\rho^l(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(x)\,U^{ac}(z)\,\rho^c(x')\,\rho^b(x)
\end{aligned}
$$

$$
\begin{aligned}
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(z)\,\rho^c(x')\,\rho^b(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{be}(z)\,U^{ch}(w)\,\rho^l(x')\,\rho^d(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{be}(z)\,U^{ch}(w)\,\rho^l(x')\,\rho^d(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{be}(z)\,U^{ch}(w')\,\rho^d(x')\,\rho^l(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{be}(z)\,U^{ch}(w')\,\rho^d(x')\,\rho^l(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(z)\,\rho^c(x')\,\rho^b(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(z)\,\rho^c(x')\,\rho^b(x)\Bigg]
\end{aligned}
\tag{4.2}
$$

Group I, Row V, part 2b finite: the two endpoint terms (B_2, B_4).

$$
\begin{aligned}
\left.\frac{d^3N}{d^3k}\right|_{G1\text{-}V\ 2b\,\mathrm{end}} = \mathcal{N}\log\frac{\vee}{\Lambda}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(x')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(w')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(x)\,U^{ac}(z)\,\rho^c(x')\,\rho^b(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(z)\,\rho^c(x')\,\rho^b(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(z)\,\rho^c(x')\,\rho^b(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(z)\,\rho^c(x')\,\rho^b(x)\Bigg]
\end{aligned}
\tag{4.3}
$$

Group I, Row V, part 2b UV (bracket written as $\frac{1}{\pi}\int_x[K(z,z,x)-K(y,z,x)]$).

$$\begin{aligned}
\left.\frac{d^3N}{d^3k}\right|_{G1\text{-}V\ 2b\,\mathrm{UV}} = \mathcal{N}\log\frac{\vee}{\Lambda}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[& \\
& -2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
& +2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
& +2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
& -2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
& -2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
& +2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
& +2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
& -2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
& +2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
& -2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
& -2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
& +2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
& +2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
& -2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
& -2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
& +2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\Bigg]
\end{aligned} \tag{4.4}$$

Group I, Row V, part 2b finite: the counterterm.

$$\begin{aligned}
\left.\frac{d^3N}{d^3k}\right|_{G1\text{-}V\ 2b\,\mathrm{CT}} = \mathcal{N}\log\frac{\vee}{\Lambda}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[& \\
& +2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
& -2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
& -2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
& +2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
& +2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
& -2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
& -2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
& +2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
& -2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
& +2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
& +2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
& -2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
& -2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
& +2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
& +2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
& -2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\Bigg]
\end{aligned} \tag{4.5}$$

Group II, Row I.

$$\left.\frac{d^3N}{d^3k}\right|_{G2\text{-}I} = \mathcal{N}\log\frac{\vee}{\Lambda}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[$$

$$
\begin{aligned}
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^{c}(x')\,\rho^{e}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x')\,U^{bc}(x)\,\rho^{a}(x')\,\rho^{c}(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^{c}(x')\,\rho^{e}(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^{l}(x')\,\rho^{h}(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(x')\,U^{bc}(x)\,\rho^{a}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^{c}(x')\,\rho^{e}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^{l}(x')\,\rho^{h}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(x')\,U^{bc}(x)\,\rho^{a}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\;f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^{c}(x')\,\rho^{e}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\;f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^{l}(x')\,\rho^{h}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\;U^{ab}(x')\,U^{bc}(x)\,\rho^{a}(x')\,\rho^{c}(x)\Big]
\end{aligned}
\tag{4.6}
$$

Group II, Row IV.

$$
\begin{aligned}
\left.\frac{d^3N}{d^3k}\right|_{G2\text{-}IV} = \mathcal{N}\log\frac{\vee}{\Lambda}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[&\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;U^{ab}(x')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;U^{ab}(w')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\;U^{ab}(x)\,U^{ac}(w')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\;U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\;U^{ab}(w)\,U^{ac}(w')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\;U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(x')\,U^{bc}(x)\,\rho^{a}(x')\,\rho^{c}(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(w')\,U^{cb}(w)\,\rho^{a}(x')\,\rho^{c}(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(w')\,U^{cb}(x)\,\rho^{a}(x')\,\rho^{c}(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(x')\,U^{cb}(w)\,\rho^{a}(x')\,\rho^{c}(x)\Big]
\end{aligned}
\tag{4.7}
$$

Group III, Row III, second part.

$$
\begin{aligned}
\left.\frac{d^3N}{d^3k}\right|_{G3\text{-}III(2)} = \mathcal{N}\log\frac{\vee}{\Lambda}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[&\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^{l}(x')\,\rho^{h}(x)\\
&-4\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(x')\,U^{bc}(x)\,\rho^{a}(x')\,\rho^{c}(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(w')\,U^{cb}(x)\,\rho^{a}(x')\,\rho^{c}(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(x')\,U^{cb}(w)\,\rho^{a}(x')\,\rho^{c}(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{deh}\,U^{la}(w')\,U^{be}(z)\,U^{cd}(w)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{deh}\,U^{la}(w)\,U^{be}(z)\,U^{cd}(w')\,\rho^{h}(x')\,\rho^{l}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^{l}(x')\,\rho^{h}(x)
\end{aligned}
$$

$$
\begin{aligned}
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{deh}\,U^{la}(w')\,U^{be}(z)\,U^{cd}(w)\,\rho^l(x')\,\rho^h(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(w')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\;f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\;U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\;f^{abc}\,f^{deh}\,U^{la}(w)\,U^{be}(z)\,U^{cd}(w')\,\rho^h(x')\,\rho^l(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\;U^{ab}(w)\,U^{bc}(x')\,\rho^c(x')\,\rho^a(x)\Big]
\end{aligned}
\tag{4.8}
$$

4.2 From four ρ's: the two-ρ lists of `4rho.tex` (missing in `Evolution_terms.tex`, M1)

These are the two-ρ terms of `4rho.tex` ("Two ρ contribution", $dN^{(1)}$, $dN^{(2)}$ with c.c.), with the two fixes found earlier:

- **F1:** in the first part, drop $-8N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-w)\,K(y,x',z)$;
- **F2:** in the second part, $dN^{(1)}$, term 2 has one index four times. The correct term is

$$3f^{abc}f^{deg}\rho^{b'}(x')\rho^c(y)U^{ag}(y)U^{db'}(w')U^{eb}(z).$$

Collected (added):

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|_{4\rho\text{-lists}} = \mathcal{N}\log\frac{\vee}{\Lambda}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{bl}(w)\,U^{dh}(x)\,U^{ml}(w')\,\rho^m(x')\,\rho^e(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{bl}(x)\,U^{dh}(x)\,U^{ml}(w')\,\rho^m(x')\,\rho^e(x)\\
&-3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{ah}(w)\,U^{bl}(x)\,U^{dl}(z)\,U^{mh}(w')\,\rho^m(x')\,\rho^e(x)\\
&+3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{dl}(w)\,U^{lm}(w')\,U^{hb}(x)\,\rho^m(x')\,\rho^e(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hd}(w)\,U^{hl}(x')\,U^{ma}(z)\,U^{mb}(x)\,\rho^l(x')\,\rho^e(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{bl}(x)\,U^{dh}(x)\,U^{ml}(x')\,\rho^m(x')\,\rho^e(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{hl}(w')\,U^{ma}(z)\,U^{mb}(x)\,\rho^l(x')\,\rho^e(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{hl}(x')\,U^{ma}(z)\,U^{mb}(x)\,\rho^l(x')\,\rho^e(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{ha}(w)\,U^{hl}(x')\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hb}(w)\,U^{hl}(x')\,U^{ma}(x)\,U^{md}(z)\,\rho^l(x')\,\rho^e(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{deh}\,U^{ad}(z)\,U^{bh}(x)\,U^{el}(x')\,\rho^l(x')\,\rho^c(x)\\
&+3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{deh}\,U^{ah}(x)\,U^{dl}(w')\,U^{eb}(z)\,\rho^l(x')\,\rho^c(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{cde}\,U^{hb}(x')\,U^{hl}(w)\,U^{ma}(z)\,U^{md}(x')\,\rho^l(x)\,\rho^e(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{cde}\,U^{hb}(x')\,U^{hl}(x)\,U^{ma}(z)\,U^{md}(x')\,\rho^l(x)\,\rho^e(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{hl}(w)\,U^{ma}(z)\,U^{mb}(x')\,\rho^l(x)\,\rho^e(x')\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{hl}(x)\,U^{ma}(z)\,U^{mb}(x')\,\rho^l(x)\,\rho^e(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^l(x')\,\rho^h(x)\\
&+\tfrac{3}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{eb}(w)\,U^{lh}(x')\,U^{da}(x)\,\rho^l(x')\,\rho^c(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{ade}\,U^{hb}(w)\,U^{hd}(x')\,\rho^e(x')\,\rho^c(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{ade}\,U^{hb}(x)\,U^{hd}(x')\,\rho^e(x')\,\rho^c(x)
\end{aligned}
$$

$$
\begin{aligned}
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dae}\,U^{hb}(w')\,U^{hd}(x)\,\rho^{e}(x)\,\rho^{c}(x')\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{eb}(x')\,U^{lh}(x)\,U^{da}(z)\,\rho^{l}(x)\,\rho^{c}(x')\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{da}(z)\,U^{lc}(x')\,U^{be}(x)\,\rho^{h}(x)\,\rho^{l}(x')\Big]
\end{aligned}
\tag{4.9}
$$

4.3 Your Sec. 2 (from four and three ρ's), coloured

Verbatim from `Evolution_terms.tex`; only the colours are added. Every row is correct (Sec. 8).

From $4\,\rho's$

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{\vee}{\Lambda}\Big]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\Big[\\
&-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{bc}(w)\,\rho^{a}(x')\,\rho^{c}(x)\\
&+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w)\,U^{ca}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{bc}(x)\,\rho^{a}(x')\,\rho^{c}(x)\\
&+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w')\,U^{ca}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w')\,U^{cb}(w)\,\rho^{a}(x')\,\rho^{c}(x)\\
&-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{bc}(x')\,\rho^{a}(x)\,\rho^{c}(x')\\
&+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{dh}(w')\,U^{hb}(w)\,\rho^{e}(x')\,\rho^{c}(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(w)\,U^{dh}(x')\,\rho^{e}(x')\,\rho^{c}(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dae}\,U^{hb}(w')\,U^{dh}(x)\,\rho^{c}(x')\,\rho^{e}(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{eb}(w')\,U^{lc}(x)\,U^{ad}(x')\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{lc}(x')\,U^{eb}(w)\,U^{ad}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{dh}(w')\,U^{hb}(w)\,\rho^{a}(x)\,\rho^{e}(x')\\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hb}(w)\,U^{dh}(x')\,\rho^{a}(x)\,\rho^{e}(x')\\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hb}(w')\,U^{dh}(x')\,\rho^{a}(x)\,\rho^{e}(x')\\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{bh}(w')\,U^{dh}(w)\,\rho^{a}(x)\,\rho^{e}(x')\\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{eb}(w')\,U^{dc}(x')\,U^{al}(x)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ec}(x')\,U^{db}(w)\,U^{al}(x)\,\rho^{h}(x')\,\rho^{l}(x)\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\Big]
\end{aligned}
\tag{4.10}
$$

From three ρ's

Group I (with c.c.)

Row I Only 4 ρ's

Row II

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{\Lambda}\right]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Bigg[$$
$$-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$$
$$+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^l(x)\,\rho^h(x')$$
$$-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(x')\,U^{ce}(x)\,\rho^l(x')\,\rho^h(x)$$
$$+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ah}(w')\,\rho^b(x')\,\rho^e(x)$$
$$+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$$
$$-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ah}(w')\,\rho^b(x')\,\rho^e(x)$$
$$+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(x')\,U^{ce}(x)\,\rho^l(x')\,\rho^h(x)$$
$$-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ah}(w)\,\rho^b(x)\,\rho^e(x')$$
$$+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$$
$$+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ah}(w)\,\rho^b(x)\,\rho^e(x')$$
$$-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^l(x)\,\rho^h(x')$$
$$+2\,N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$$
$$+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x)$$
$$-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(x')\,U^{cl}(x)\,\rho^l(x)\,\rho^h(x')$$
$$-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{bde}\,U^{hd}(x')\,U^{ah}(w)\,\rho^e(x')\,\rho^c(x)$$
$$-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$$
$$+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ah}(w)\,\rho^e(x')\,\rho^c(x)$$
$$-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x)$$
$$+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ah}(w')\,\rho^c(x)\,\rho^e(x')$$
$$-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$$
$$-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dbe}\,U^{hd}(w)\,U^{ah}(w')\,\rho^c(x)\,\rho^e(x')$$
$$+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(x')\,U^{cl}(x)\,\rho^l(x)\,\rho^h(x')\Bigg] \tag{4.11}$$

Row III We write it in two parts:

$$\frac{d^3N_{1\,\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{\Lambda}\right]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Bigg[$$
$$+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x)$$
$$-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$$
$$-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x)$$
$$+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$$
$$-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x)$$
$$-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$$
$$+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x)$$
$$+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$$
$$+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')$$
$$+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$$
$$-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')$$

$$
\begin{aligned}
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(z)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^{l}(x)\,\rho^{h}(x')\Bigg]
\end{aligned}
\tag{4.12}
$$

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{\Lambda}\right]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\Bigg[\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(w)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(w)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x)\,U^{bl}(w')\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x)\,U^{bl}(x')\,\rho^{l}(x')\,\rho^{h}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(w)\,\rho^{h}(x')\,\rho^{l}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(w')\,\rho^{h}(x')\,\rho^{l}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(w')\,\rho^{h}(x')\,\rho^{l}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(w)\,\rho^{l}(x)\,\rho^{h}(x')\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(w)\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{cl}(x)\,U^{be}(w')\,\rho^{l}(x)\,\rho^{h}(x')\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x)\,U^{be}(w')\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x)\,U^{be}(x')\,\rho^{l}(x)\,\rho^{h}(x')\Bigg]
\end{aligned}
\tag{4.13}
$$

Row IV We write it in two parts:

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{\Lambda}\right]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\Bigg[\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ah}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)
\end{aligned}
$$

$$
\begin{aligned}
&+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ah}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)\\
&-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ah}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)\\
&+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{bc}(x')\,\rho^c(x')\,\rho^a(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ah}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)\\
&-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{bc}(x')\,\rho^c(x')\,\rho^a(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w)\,U^{db}(z)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ca}(x')\,\rho^c(x')\,\rho^b(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{le}(w')\,U^{al}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^h(x')\,\rho^c(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ea}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{eb}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{al}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^c(x')\,\rho^h(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{da}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^l(x')\,\rho^h(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ah}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ah}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{eb}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{al}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^c(x')\,\rho^h(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{be}(x)\,U^{ad}(w)\,U^{cl}(x')\,\rho^l(x')\,\rho^h(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w)\,U^{db}(z)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{le}(w')\,U^{al}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^h(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{cb}(x')\,\rho^c(x')\,\rho^a(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ea}(w')\,U^{bd}(x')\,U^{lc}(x)\,\rho^h(x')\,\rho^l(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ah}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ah}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ah}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ca}(x')\,\rho^b(x)\,\rho^c(x')\Bigg]
\end{aligned}
\tag{4.14}
$$

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{\Lambda}\right]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Bigg[\\
-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ah}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')\\
+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ea}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^l(x)\,\rho^h(x')
\end{aligned}
$$

$$
\begin{aligned}
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ah}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')\\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ah}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')\\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{da}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^l(x)\,\rho^h(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ah}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ah}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')\\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ca}(x')\,\rho^b(x)\,\rho^c(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ea}(w')\,U^{bl}(x)\,U^{dc}(x')\,\rho^l(x)\,\rho^h(x')\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ah}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ah}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{bl}(x)\,U^{ad}(w)\,U^{ce}(x')\,\rho^l(x)\,\rho^h(x')\\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\Bigg]
\end{aligned}
\tag{4.15}
$$

Row V

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{\Lambda}\right]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\Bigg[\\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{be}(z)\,U^{cl}(w')\,\rho^l(x')\,\rho^h(x)\\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(z)\,U^{cl}(w')\,\rho^l(x')\,\rho^h(x)\\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(x)\,U^{cl}(x')\,\rho^l(x')\,\rho^h(x)\\
-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{bd}(z)\,U^{cl}(w)\,\rho^l(x)\,\rho^h(x')\\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(z)\,U^{cl}(w)\,\rho^l(x)\,\rho^h(x')\\
-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^l(x)\,\rho^h(x')\\
-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{bl}(z)\,U^{cd}(w)\,\rho^h(x')\,\rho^l(x)\\
+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(z)\,U^{cd}(w)\,\rho^h(x')\,\rho^l(x)\\
+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^h(x')\,\rho^l(x)\\
+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{bl}(z)\,U^{ce}(w')\,\rho^l(x)\,\rho^h(x')\\
-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(z)\,U^{ce}(w')\,\rho^l(x)\,\rho^h(x')\\
+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(x)\,U^{ce}(x')\,\rho^l(x)\,\rho^h(x')\\
+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\Bigg]
\end{aligned}
\tag{4.16}
$$

Group II

Row I

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{\Lambda}\right]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\Bigg[\\
\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ca}(w)\,\rho^b(x')\,\rho^c(x)\\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{be}(z)\,U^{al}(w')\,U^{cd}(x)\,\rho^l(x')\,\rho^h(x)\\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{bm}(z)\,U^{md}(x)\,U^{ah}(w)\,\rho^l(x')\,\rho^e(x)\\
+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w')\,U^{bc}(w)\,\rho^c(x)\,\rho^a(x')
\end{aligned}
$$

$$
\begin{aligned}
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{ah}(w')\,U^{bl}(z)\,U^{hm}(x)\,U^{ld}(x')\,\rho^m(x)\,\rho^e(x')\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{bd}(z)\,U^{ce}(x')\,U^{al}(w)\,\rho^l(x)\,\rho^h(x')\\
&+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ca}(w)\,\rho^b(x')\,\rho^c(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{eb}(z)\,U^{cd}(x)\,\rho^l(x')\,\rho^h(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{bm}(z)\,U^{md}(x)\,U^{ha}(w)\,\rho^l(x')\,\rho^e(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w')\,U^{bc}(w)\,\rho^c(x)\,\rho^a(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{ah}(w')\,U^{bl}(z)\,U^{hm}(x)\,U^{ld}(x')\,\rho^m(x)\,\rho^e(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ce}(x')\,U^{db}(z)\,U^{al}(w)\,\rho^l(x)\,\rho^h(x')\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ah}(w)\,\rho^b(x)\,\rho^e(x')\\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{bl}(z)\,U^{ae}(w')\,U^{cd}(x')\,\rho^l(x)\,\rho^h(x')\\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{bl}(z)\,U^{lm}(x)\,U^{ah}(w)\,\rho^m(x)\,\rho^e(x')\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{ah}(w')\,U^{hd}(w)\,\rho^e(x')\,\rho^b(x)\\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{ah}(w')\,U^{bl}(z)\,U^{hd}(x')\,U^{lm}(x)\,\rho^e(x')\,\rho^m(x)\\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{bl}(z)\,U^{ce}(x')\,U^{ad}(w)\,\rho^h(x')\,\rho^l(x)\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ah}(w)\,\rho^b(x)\,\rho^e(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{lb}(z)\,U^{cd}(x')\,\rho^l(x)\,\rho^h(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{bl}(z)\,U^{lm}(x)\,U^{ha}(w)\,\rho^m(x)\,\rho^e(x')\\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ca}(x')\,\rho^b(x)\,\rho^c(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{ah}(w')\,U^{hd}(w)\,\rho^e(x')\,\rho^b(x)\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{ah}(w')\,U^{bl}(z)\,U^{hd}(x')\,U^{lm}(x)\,\rho^e(x')\,\rho^m(x)\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ce}(x')\,U^{lb}(z)\,U^{ad}(w)\,\rho^h(x')\,\rho^l(x)\\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\Big]
\end{aligned}
\tag{4.17}
$$

Row II Only 4 ρ's

Row III Only 4 ρ's

Row IV Only 2 ρ's

Group III (with c.c.)

Row I We write it in two parts:

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{\Lambda}\right]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Big[\\
-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{md}(x)\,U^{mb}(z)\,U^{ah}(w)\,\rho^l(x')\,\rho^e(x)\\
-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{bh}(z)\,U^{hd}(x)\,U^{lm}(w')\,U^{al}(w)\,\rho^m(x')\,\rho^e(x)
\end{aligned}
$$

$$
\begin{aligned}
&+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{md}(x')\,U^{mb}(z)\,U^{ah}(w')\,\rho^e(x')\,\rho^l(x)\\
&-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{bh}(z)\,U^{hd}(x')\,U^{lm}(w)\,U^{la}(w')\,\rho^e(x')\,\rho^m(x)\\
&+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{db}(z)\,U^{ae}(w)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{bl}(z)\,U^{ld}(x')\,U^{me}(w')\,U^{am}(w)\,\rho^h(x')\,\rho^c(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{ad}(w')\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{bl}(z)\,U^{le}(x)\,U^{md}(w)\,U^{ma}(w')\,\rho^c(x')\,\rho^h(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{be}(x)\,U^{ad}(w)\,U^{cl}(x')\,\rho^l(x')\,\rho^h(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{md}(x)\,U^{mb}(z)\,U^{ah}(w)\,\rho^l(x')\,\rho^e(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{bh}(z)\,U^{hd}(x)\,U^{lm}(w')\,U^{al}(w)\,\rho^m(x')\,\rho^e(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{ad}(w')\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{le}(x)\,U^{md}(w)\,U^{lb}(z)\,U^{ma}(w')\,\rho^c(x')\,\rho^h(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{be}(x)\,U^{ad}(w)\,U^{cl}(x')\,\rho^l(x')\,\rho^h(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{db}(z)\,U^{ae}(w)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{bl}(z)\,U^{ld}(x')\,U^{me}(w')\,U^{am}(w)\,\rho^h(x')\,\rho^c(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{md}(x')\,U^{mb}(z)\,U^{ah}(w')\,\rho^e(x')\,\rho^l(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{lm}(w)\,U^{hb}(z)\,U^{la}(w')\,\rho^e(x')\,\rho^m(x)\Big]
\end{aligned}
\tag{4.18}
$$

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{\Lambda}\right]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Big[\\
-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{lm}(x)\,U^{lb}(z)\,U^{ah}(w)\,\rho^m(x)\,\rho^e(x')\\
+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{bh}(z)\,U^{hl}(x)\,U^{md}(w')\,U^{am}(w)\,\rho^l(x)\,\rho^e(x')\\
-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^l(x)\,\rho^h(x')\\
+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{lm}(x)\,U^{lb}(z)\,U^{ah}(w')\,\rho^m(x)\,\rho^e(x')\\
-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{bh}(z)\,U^{hl}(x)\,U^{md}(w)\,U^{ma}(w')\,\rho^l(x)\,\rho^e(x')\\
+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{bl}(x)\,U^{ad}(w)\,U^{ce}(x')\,\rho^l(x)\,\rho^h(x')
\end{aligned}
$$

$$
\begin{aligned}
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{lm}(x)\,U^{lb}(z)\,U^{ah}(w)\,\rho^{m}(x)\,\rho^{e}(x')\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{bh}(z)\,U^{hl}(x)\,U^{md}(w')\,U^{am}(w)\,\rho^{l}(x)\,\rho^{e}(x')\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{lm}(x)\,U^{lb}(z)\,U^{ah}(w')\,\rho^{m}(x)\,\rho^{e}(x')\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{md}(w)\,U^{hb}(z)\,U^{ma}(w')\,\rho^{l}(x)\,\rho^{e}(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{bl}(x)\,U^{ad}(w)\,U^{ce}(x')\,\rho^{l}(x)\,\rho^{h}(x')\Big]
\end{aligned}
\tag{4.19}
$$

Row II We write it in two parts:

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{\vee}{\Lambda}\Big]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Big[\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{lm}(x')\,U^{bh}(z)\,U^{al}(w)\,\rho^{m}(x')\,\rho^{e}(x)\\
&-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ca}(w)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{eb}(z)\,U^{cd}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{lm}(x)\,U^{bh}(z)\,U^{al}(w')\,\rho^{m}(x)\,\rho^{e}(x')\\
&-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ca}(w')\,\rho^{b}(x)\,\rho^{c}(x')\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{db}(z)\,U^{ce}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{be}(z)\,U^{ad}(w)\,U^{hl}(x')\,\rho^{l}(x')\,\rho^{c}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ah}(w)\,\rho^{e}(x')\,\rho^{c}(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{eb}(z)\,U^{cl}(x)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{bd}(z)\,U^{ae}(w')\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{bde}\,U^{hd}(x)\,U^{ah}(w')\,\rho^{c}(x')\,\rho^{e}(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(x)\,U^{db}(z)\,U^{cl}(x')\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{lm}(x')\,U^{bh}(z)\,U^{la}(w)\,\rho^{m}(x')\,\rho^{e}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ca}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{eb}(z)\,U^{cd}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{bd}(z)\,U^{ea}(w')\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{bde}\,U^{hd}(x)\,U^{ah}(w')\,\rho^{c}(x')\,\rho^{e}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(x)\,U^{db}(z)\,U^{cl}(x')\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{be}(z)\,U^{da}(w)\,U^{hl}(x')\,\rho^{l}(x')\,\rho^{c}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ha}(w)\,\rho^{e}(x')\,\rho^{c}(x)
\end{aligned}
$$

$$
\begin{aligned}
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{eb}(z)\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{lm}(x)\,U^{bh}(z)\,U^{la}(w')\,\rho^m(x)\,\rho^e(x')\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ca}(w')\,\rho^b(x)\,\rho^c(x')\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{db}(z)\,U^{ce}(x')\,\rho^l(x)\,\rho^h(x')\\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hl}(x)\,U^{md}(x')\,U^{bh}(z)\,U^{am}(w)\,\rho^l(x)\,\rho^e(x')\\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ah}(w)\,\rho^b(x)\,\rho^e(x')\Big]
\end{aligned}
\tag{4.20}
$$

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{\Lambda}\right]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Big[\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(z)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x')\\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{md}(x')\,U^{bh}(z)\,U^{am}(w')\,\rho^e(x')\,\rho^l(x)\\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ah}(w')\,\rho^e(x')\,\rho^b(x)\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hl}(x)\,U^{md}(x')\,U^{bh}(z)\,U^{ma}(w)\,\rho^l(x)\,\rho^e(x')\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')\\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(z)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{md}(x')\,U^{bh}(z)\,U^{ma}(w')\,\rho^e(x')\,\rho^l(x)\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ah}(w')\,\rho^e(x')\,\rho^b(x)\\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(z)\,U^{bc}(x')\,\rho^c(x')\,\rho^a(x)\Big]
\end{aligned}
\tag{4.21}
$$

Row III

$$
\begin{aligned}
\frac{d^3N_{\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{\Lambda}\right]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Big[\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(z)\,U^{cl}(w')\,\rho^l(x')\,\rho^h(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^l(x')\,\rho^h(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{ad}(x)\,U^{cl}(w')\,\rho^l(x')\,\rho^h(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(z)\,U^{cl}(w)\,\rho^l(x)\,\rho^h(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^l(x)\,\rho^h(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{bd}(z)\,U^{ae}(x')\,U^{cl}(w)\,\rho^l(x)\,\rho^h(x')\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(z)\,U^{ce}(w')\,\rho^l(x)\,\rho^h(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^l(x)\,\rho^h(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{lb}(z)\,U^{ad}(x')\,U^{ce}(w')\,\rho^l(x)\,\rho^h(x')\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(z)\,U^{cd}(w)\,\rho^h(x')\,\rho^l(x)\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^h(x')\,\rho^l(x)\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{bl}(z)\,U^{ae}(x')\,U^{cd}(w)\,\rho^h(x')\,\rho^l(x)\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\Big]
\end{aligned}
\tag{4.22}
$$

Row IV Only 4 ρ's

Row V

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{\Lambda}\right]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Bigg[\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(x)\,U^{cl}(x')\,\rho^l(x')\,\rho^h(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(w')\,U^{be}(x)\,\rho^l(x')\,\rho^h(x) \\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{cd}(w)\,U^{be}(x)\,\rho^l(x')\,\rho^h(x) \\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{cd}(w)\,U^{be}(x)\,\rho^l(x')\,\rho^h(x) \\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{ce}(w')\,U^{bd}(x')\,\rho^h(x')\,\rho^l(x) \\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{ce}(w')\,U^{bd}(x')\,\rho^h(x')\,\rho^l(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(w)\,U^{bd}(x')\,\rho^h(x')\,\rho^l(x) \\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(x)\,U^{ce}(x')\,\rho^l(x)\,\rho^h(x') \\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(w')\,U^{bl}(x)\,\rho^l(x)\,\rho^h(x') \\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x') \\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{ce}(w')\,U^{bl}(x)\,\rho^l(x)\,\rho^h(x') \\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^l(x)\,\rho^h(x') \\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(w)\,U^{bl}(x)\,\rho^l(x)\,\rho^h(x') \\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x') \\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cd}(w)\,U^{bl}(x)\,\rho^l(x)\,\rho^h(x')\Bigg]
\end{aligned}
\tag{4.23}
$$

Row VI

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{\Lambda}\right]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Bigg[\\
&\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^l(x')\,\rho^h(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(z)\,U^{cl}(x')\,\rho^l(x')\,\rho^h(x) \\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{eb}(z)\,\rho^l(x')\,\rho^h(x) \\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{bd}(z)\,U^{ce}(x)\,\rho^l(x')\,\rho^h(x) \\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x) \\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{be}(z)\,U^{cd}(x')\,\rho^h(x')\,\rho^l(x) \\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{eb}(z)\,\rho^h(x')\,\rho^l(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^l(x)\,\rho^h(x') \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(z)\,U^{cl}(x)\,\rho^l(x)\,\rho^h(x') \\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')
\end{aligned}
$$

$$
\begin{aligned}
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(z)\,U^{ce}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(z)\,U^{bc}(x')\,\rho^{a}(x)\,\rho^{c}(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(z)\,U^{cd}(x')\,\rho^{h}(x')\,\rho^{l}(x)\\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(z)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\Bigg]
\end{aligned}
\tag{4.24}
$$

4.4 The sum

Kernel structure by kernel structure, the colour structures of all terms above are reduced to an independent basis. The sum minus JIMWLK⊗LO is then:

kernel structure	class	independent colour structures	differences
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$	charge–charge	12	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$	charge–charge	10	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	charge–charge	13	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w,w,z)$	JIMWLK only	10	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w',w',z)$	JIMWLK only	10	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w,w',z)$	JIMWLK only	9	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	JIMWLK only	14	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	JIMWLK only	12	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$	JIMWLK only	12	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$	JIMWLK only	14	0
$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	charge–charge	14	0
$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	JIMWLK only	12	0
$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	JIMWLK only	12	0

Without the `4rho.tex` lists, 18 of 152 colour structures would differ (in $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$, $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$, $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$). The sum, collected, is JIMWLK⊗LO; in this form it reads

$$
\begin{aligned}
\left.\frac{d^3N}{d^3k}\right|_{\text{JIMWLK}\otimes\text{LO}} = \mathcal{N}\log\frac{\vee}{\Lambda}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[
&+2\,N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(x-z)^2}\;U^{ab}(x)\,U^{ac}(w')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-2\,N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(x-z)^2}\;U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{bl}(x)\,U^{dh}(x)\,U^{ml}(w')\,\rho^{m}(x')\,\rho^{e}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{bl}(x)\,U^{dh}(x)\,U^{ml}(x')\,\rho^{m}(x')\,\rho^{e}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{hl}(w')\,U^{ma}(z)\,U^{mb}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{hl}(x')\,U^{ma}(z)\,U^{mb}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{deh}\,U^{ad}(z)\,U^{bh}(x)\,U^{el}(x')\,\rho^{l}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{deh}\,U^{ah}(x)\,U^{dl}(w')\,U^{eb}(z)\,\rho^{l}(x')\,\rho^{c}(x)\\
&+2\,N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(x'-z)^2}\;U^{ab}(x')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-2\,N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(x'-z)^2}\;U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{cde}\,U^{hb}(x')\,U^{hl}(w)\,U^{ma}(z)\,U^{md}(x')\,\rho^{l}(x)\,\rho^{e}(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{cde}\,U^{hb}(x')\,U^{hl}(x)\,U^{ma}(z)\,U^{md}(x')\,\rho^{l}(x)\,\rho^{e}(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{hl}(w)\,U^{ma}(z)\,U^{mb}(x')\,\rho^{l}(x)\,\rho^{e}(x')\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{hl}(x)\,U^{ma}(z)\,U^{mb}(x')\,\rho^{l}(x)\,\rho^{e}(x')\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+2\,N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x')\,U^{bc}(x)\,\rho^{a}(x')\,\rho^{c}(x)
\end{aligned}
$$

$$
\begin{aligned}
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{ade}\,U^{hb}(x)\,U^{hd}(x')\,\rho^{e}(x')\,\rho^{c}(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{eb}(x')\,U^{lh}(x)\,U^{da}(z)\,\rho^{l}(x)\,\rho^{c}(x')\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{da}(z)\,U^{lc}(x')\,U^{be}(x)\,\rho^{h}(x)\,\rho^{l}(x')\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;U^{ab}(x')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;U^{ab}(w')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(x')\,U^{be}(z)\,U^{ch}(w)\,\rho^{l}(x')\,\rho^{d}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(w')\,U^{be}(z)\,U^{ch}(w)\,\rho^{l}(x')\,\rho^{d}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;f^{abc}\,f^{dea}\,U^{hd}(z)\,U^{he}(w)\,U^{lm}(w')\,U^{lb}(w)\,\rho^{m}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;f^{abc}\,f^{deb}\,U^{hd}(z)\,U^{ha}(w)\,U^{lm}(w')\,U^{le}(w)\,\rho^{m}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;f^{abc}\,f^{dea}\,U^{hd}(z)\,U^{he}(w)\,U^{lm}(x')\,U^{lb}(w)\,\rho^{m}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;f^{abc}\,f^{deb}\,U^{hd}(z)\,U^{ha}(w)\,U^{lm}(x')\,U^{le}(w)\,\rho^{m}(x')\,\rho^{c}(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\;U^{ab}(x)\,U^{ac}(w')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\;U^{ab}(w)\,U^{ac}(w')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(x)\,U^{be}(z)\,U^{ch}(w')\,\rho^{d}(x')\,\rho^{l}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(w)\,U^{be}(z)\,U^{ch}(w')\,\rho^{d}(x')\,\rho^{l}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\;f^{abc}\,f^{dea}\,U^{hd}(z)\,U^{he}(w')\,U^{lb}(w')\,U^{lm}(w)\,\rho^{c}(x')\,\rho^{m}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\;f^{abc}\,f^{deb}\,U^{hd}(z)\,U^{ha}(w')\,U^{le}(w')\,U^{lm}(w)\,\rho^{c}(x')\,\rho^{m}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\;f^{abc}\,f^{dea}\,U^{hd}(z)\,U^{he}(w')\,U^{lb}(w')\,U^{lm}(x)\,\rho^{c}(x')\,\rho^{m}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\;f^{abc}\,f^{deb}\,U^{hd}(z)\,U^{ha}(w')\,U^{le}(w')\,U^{lm}(x)\,\rho^{c}(x')\,\rho^{m}(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^{c}(x')\,\rho^{e}(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(w')\,U^{cb}(w)\,\rho^{a}(x')\,\rho^{c}(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{deh}\,U^{la}(w')\,U^{be}(z)\,U^{cd}(w)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{deh}\,U^{la}(w)\,U^{be}(z)\,U^{cd}(w')\,\rho^{h}(x')\,\rho^{l}(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(w')\,U^{bc}(x)\,\rho^{a}(x')\,\rho^{c}(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ah}(w')\,\rho^{b}(x')\,\rho^{e}(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{be}(z)\,U^{cl}(w')\,\rho^{l}(x')\,\rho^{h}(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^{l}(x')\,\rho^{h}(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\;U^{ab}(w)\,U^{bc}(x')\,\rho^{c}(x')\,\rho^{a}(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\;f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ah}(w)\,\rho^{b}(x)\,\rho^{e}(x')\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\;f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{bd}(z)\,U^{cl}(w)\,\rho^{l}(x)\,\rho^{h}(x')
\end{aligned}
\tag{4.25}
$$

This is $\mathcal{N}\log\frac{\vee}{\Lambda}\times\mathcal{J}$, Eq. (3.1), before the charge average. Its terms (green / purple): 28 / 20.

5 BFKL: the coefficient of $\log\vee$

For every row, the coefficient of $\log\vee$ is the hard end $\mathfrak{P}$ of Sec. 2. It equals the coefficient of $\log\frac{\vee}{\Lambda}$ plus that of $\log\frac{\vee}{k^+}$.

- It is zero for Group I Row V, for $\mathcal{D}$ and for the genuine Group I terms.
- For the four-ρ remainders and the `4rho.tex` lists it equals their $\log\frac{\vee}{\Lambda}$ terms.
- For Group I Row III it needs the sign of the $p^+ > k^+ + \Lambda$ line (fix P1). This is the sign now in your `Group_I.tex`, Eq. (1.10).

The two-ρ terms of the coefficient of $\log\vee$, row by row (orange / purple: 168 / 162):

Group I, Row II (three ρ; the α part, fix F4).

$$\frac{d^3N}{d^3k}\Big|^{\log\vee}_{I.II} = \mathcal{N}\log\vee\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[$$
$$-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')$$
$$+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,\rho^b(x')\,\rho^e(x)$$
$$+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')$$
$$-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{bde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^e(x')\,\rho^c(x)$$
$$+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ha}(w')\,\rho^c(x)\,\rho^e(x')$$
$$+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^e(x')\,\rho^c(x)$$
$$-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dbe}\,U^{hd}(w)\,U^{ha}(w')\,\rho^c(x)\,\rho^e(x')\Big] \tag{5.1}$$

Group I, Row III (three ρ; with the sign of $p^+ > k^+$, fix P1).

$$\frac{d^3N}{d^3k}\Big|^{\log\vee}_{I.III} = \mathcal{N}\log\vee\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[$$
$$-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x)$$
$$+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$$
$$+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x)$$
$$-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$$
$$+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x)$$
$$+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$$
$$-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x)$$
$$-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$$
$$-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')$$
$$+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)$$
$$-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$$
$$+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')$$
$$-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)$$
$$-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^l(x)\,\rho^h(x')$$
$$+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^l(x')\,\rho^h(x)$$
$$-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(z)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$$
$$+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$$
$$+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^h(x')\,\rho^l(x)$$
$$-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(z)\,\rho^l(x)\,\rho^h(x')$$
$$-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^h(x')\,\rho^l(x)$$
$$+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^l(x)\,\rho^h(x')$$
$$-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x)$$
$$+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\;U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$$
$$+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x)$$
$$-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\;U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$$
$$-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(z)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$$
$$-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')$$
$$+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^h(x')\,\rho^l(x)$$
$$+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')$$
$$+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x)$$

$$
\begin{aligned}
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(z)\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x)\,U^{bl}(z)\,\rho^l(x')\,\rho^h(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)\Big]
\end{aligned}
\tag{5.2}
$$

Group I, Row IV (three ρ).

$$
\frac{d^3N}{d^3k}\Big|^{\log\vee}_{I.IV}=\mathcal{N}\log\vee\int_{x,x',z,w,w'}e^{-ik\cdot(w'-w)}\Big[
$$

$$
\begin{aligned}
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^l(x)\,\rho^h(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^l(x')\,\rho^h(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{le}(w')\,U^{la}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^h(x')\,\rho^c(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{la}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^c(x')\,\rho^h(x)\\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^h(x')\,\rho^l(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^l(x)\,\rho^h(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')\\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ea}(w)\,U^{db}(z)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{le}(w')\,U^{la}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^h(x')\,\rho^c(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)
\end{aligned}
$$

$$
\begin{aligned}
&- \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{da}(w')\,U^{eb}(z)\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x')\,\rho^{m}(x')\,\rho^{e}(x)\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{la}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^{c}(x')\,\rho^{h}(x)\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x')\,\rho^{m}(x')\,\rho^{e}(x)\Big]
\end{aligned}
\tag{5.3}
$$

Group II, Row I (three ρ).

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\vee}_{II.I} = \mathcal{N}\log\vee\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; U^{ab}(w')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; U^{ab}(w)\,U^{ac}(w')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^{e}(x')\,\rho^{c}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dbe}\,U^{hd}(w)\,U^{ha}(w')\,\rho^{c}(x)\,\rho^{e}(x')\\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; U^{ab}(w')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^{e}(x')\,\rho^{c}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; U^{ab}(w')\,U^{ac}(w)\,\rho^{c}(x)\,\rho^{b}(x')\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x)\,U^{bl}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x')\,\rho^{m}(x')\,\rho^{e}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{bde}\,U^{ha}(w')\,U^{hd}(w)\,\rho^{c}(x')\,\rho^{e}(x)\Big]
\end{aligned}
\tag{5.4}
$$

Group III, Row I (three ρ).

$$
\frac{d^3N}{d^3k}\Big|^{\log\vee}_{III.I} = \mathcal{N}\log\vee\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[
$$

$$
\begin{aligned}
&+\tfrac{1}{2}\, N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\, U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-\tfrac{1}{2}\, N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\, U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+\tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\, f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)\\
&-\tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\, f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)\\
&+\tfrac{1}{2}\, N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\, U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-\tfrac{1}{2}\, N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\, U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-\tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\, f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)\\
&+\tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\, f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)\\
&+N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
&+\tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')\\
&-\tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)\\
&+\tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^l(x)\,\rho^h(x')\\
&-\tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^l(x')\,\rho^h(x)\\
&-\tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, f^{abc}\,f^{deh}\,U^{le}(w')\,U^{la}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^h(x')\,\rho^c(x)\\
&+\tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{la}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^c(x')\,\rho^h(x)\\
&-N_c\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+\tfrac{1}{2}\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^h(x')\,\rho^l(x)\\
&-\tfrac{1}{2}\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^l(x)\,\rho^h(x')\\
&+\tfrac{1}{2}\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')\\
&-\tfrac{1}{2}\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')\\
&-\tfrac{1}{2}\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')\\
&+\tfrac{1}{2}\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')\\
&+N_c\, \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\, U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-N_c\, \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\, U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\, f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\, f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, f^{abc}\,f^{deh}\,U^{ea}(w)\,U^{db}(z)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, f^{abc}\,f^{deh}\,U^{le}(w')\,U^{la}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^h(x')\,\rho^c(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')\\
&+N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\, U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\, U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\, f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\, f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, f^{abc}\,f^{deh}\,U^{da}(w')\,U^{eb}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x')\,\rho^m(x')\,\rho^e(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{la}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^c(x')\,\rho^h(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x')\,\rho^m(x')\,\rho^e(x)\Big]
\end{aligned}
\tag{5.5}
$$

Group III, Row II (three ρ).

$$
\frac{d^3N}{d^3k}\bigg|^{\log\vee}_{III.II} = \mathcal{N}\log\vee\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[
$$

$$
+\tfrac{1}{2}\, N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\, U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)
$$

$$
\begin{aligned}
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;U^{ab}(x)\,U^{ac}(w')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^{b}(x)\,\rho^{e}(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,\rho^{b}(x')\,\rho^{e}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{bde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^{e}(x')\,\rho^{c}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ha}(w')\,\rho^{c}(x)\,\rho^{e}(x')\\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(z)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\;U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\;U^{ab}(x')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(z)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&-2\,N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ea}(w)\,U^{db}(z)\,U^{hl}(x')\,\rho^{l}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&-2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^{b}(x)\,\rho^{e}(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\;U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\;U^{ab}(x)\,U^{ac}(w')\,\rho^{b}(x)\,\rho^{c}(x')\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(z)\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{da}(w')\,U^{eb}(z)\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x')\,\rho^{m}(x')\,\rho^{e}(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{bde}\,U^{hd}(x)\,U^{ha}(w')\,\rho^{c}(x')\,\rho^{e}(x)\Bigg]
\end{aligned}
\tag{5.6}
$$

Group III, Row III (three ρ).

$$
\frac{d^3N}{d^3k}\bigg|^{\log\vee}_{III.III}=\mathcal{N}\log\vee\int_{x,x',z,w,w'}e^{-ik\cdot(w'-w)}\Bigg[
$$

$$
\begin{aligned}
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+N_c\;\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+N_c\;\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; U^{ab}(w')\,U^{ca}(z)\,\rho^{b}(x')\,\rho^{c}(x)\\
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&-N_c\;\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{ade}\,U^{hb}(w')\,U^{eh}(z)\,\rho^{d}(x)\,\rho^{c}(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; U^{ab}(w)\,U^{ca}(z)\,\rho^{b}(x)\,\rho^{c}(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x)\,U^{bl}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dae}\,U^{hb}(w)\,U^{eh}(z)\,\rho^{d}(x')\,\rho^{c}(x)\Big]
\end{aligned}
\tag{5.7}
$$

Group III, Row V (three ρ).

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\vee}_{III.V} = \mathcal{N}\log\vee\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[&\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-N_c\;\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+N_c\;\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+N_c\;\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(z)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&-N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(z)\,U^{ac}(x')\,\rho^{b}(x')\,\rho^{c}(x)\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x)\,U^{bl}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\Big]
\end{aligned}
\tag{5.8}
$$

Group III, Row VI (three ρ).

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\vee}_{III.VI} = \mathcal{N}\log\vee\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[&\\
&+2\;\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-N_c\;\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-N_c\;\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; U^{ab}(x')\,U^{ca}(z)\,\rho^{b}(x')\,\rho^{c}(x)
\end{aligned}
$$

$$
\begin{aligned}
&-2\,N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(z)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')\\
&+2\,N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{ade}\,U^{hb}(x')\,U^{eh}(z)\,\rho^d(x)\,\rho^c(x')\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\;U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\;U^{ab}(x)\,U^{ca}(z)\,\rho^b(x)\,\rho^c(x')\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(z)\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dae}\,U^{hb}(x)\,U^{eh}(z)\,\rho^d(x')\,\rho^c(x)\Big]
\end{aligned}
\tag{5.9}
$$

Group II, Row I (genuine two ρ).

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\vee}_{II.I^{\rm gen}} &= \mathcal{N}\log\vee\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x)\,\rho^e(x')\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\\
&-\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x')\,\rho^c(x)\,\rho^l(x')\\
&-\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)\\
&-\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x)\,\rho^l(x)\,\rho^h(x')\\
&-\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^l(x')\,\rho^h(x)\\
&+2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x')\\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ea}(w)\,U^{db}(z)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^e(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{da}(w')\,U^{eb}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{bde}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\Big]
\end{aligned}
\tag{5.10}
$$

Group II, Row IV (genuine two ρ).

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\vee}_{II.IV^{\rm gen}} &= \mathcal{N}\log\vee\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&+2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x')\\
&+2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;\rho^a(x)\,\rho^a(x')\\
&-2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(z)\,U^{cb}(x')\,\rho^a(x)\,\rho^c(x')\\
&-2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(z)\,U^{cb}(x)\,\rho^a(x')\,\rho^c(x)\Big]
\end{aligned}
\tag{5.11}
$$

Group III, Row III, second part (genuine two ρ).

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\vee}_{III.III^{\rm gen}} = \mathcal{N}\log\vee\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x')\,\rho^c(x)\,\rho^l(x') \\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x) \\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x)\,\rho^l(x)\,\rho^h(x') \\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^l(x')\,\rho^h(x) \\
&-4\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x') \\
&+2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(z)\,U^{cb}(x')\,\rho^a(x)\,\rho^c(x') \\
&+2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(z)\,U^{cb}(x)\,\rho^a(x')\,\rho^c(x) \\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dbe}\,U^{ha}(w)\,U^{eh}(z)\,\rho^d(x)\,\rho^c(x') \\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dbe}\,U^{ha}(w)\,U^{eh}(z)\,\rho^d(x')\,\rho^c(x) \\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{eh}(z)\,\rho^c(x)\,\rho^d(x') \\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{eh}(z)\,\rho^c(x')\,\rho^d(x) \\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(z)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x') \\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{ade}\,U^{hb}(w')\,U^{eh}(z)\,\rho^d(x)\,\rho^c(x') \\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(z)\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dae}\,U^{hb}(w)\,U^{eh}(z)\,\rho^d(x')\,\rho^c(x)\Big]
\end{aligned}
\tag{5.12}
$$

From four ρ's: remainders d^3N_a, d^3N_b.

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\vee}_{4\rho,\,a\text{-}b} = \mathcal{N}\log\vee\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x') \\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,\rho^b(x')\,\rho^e(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x') \\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^l(x)\,\rho^h(x') \\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^l(x')\,\rho^h(x) \\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{bde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^e(x')\,\rho^c(x) \\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ha}(w')\,\rho^c(x)\,\rho^e(x') \\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^e(x')\,\rho^c(x) \\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dbe}\,U^{hd}(w)\,U^{ha}(w')\,\rho^c(x)\,\rho^e(x') \\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^h(x')\,\rho^l(x)
\end{aligned}
$$

$$-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^{l}(x)\,\rho^{h}(x')\Big] \qquad (5.13)$$

From four ρ's: the two-ρ lists of `4rho.tex`.

$$\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\vee}_{4\rho\text{-lists}} = \mathcal{N}\log\vee\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[& \\
& -\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
& -2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
& -3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
& +2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
& +\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
& +3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
& +2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{bl}(x)\,U^{dh}(x)\,U^{ml}(w')\,\rho^{m}(x')\,\rho^{e}(x)\\
& -3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{ah}(w)\,U^{bl}(x)\,U^{dl}(z)\,U^{mh}(w')\,\rho^{m}(x')\,\rho^{e}(x)\\
& +3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{dl}(w)\,U^{lm}(w')\,U^{hb}(x)\,\rho^{m}(x')\,\rho^{e}(x)\\
& +\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hd}(w)\,U^{hl}(x')\,U^{ma}(z)\,U^{mb}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
& +2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{bl}(x)\,U^{dh}(x)\,U^{ml}(x')\,\rho^{m}(x')\,\rho^{e}(x)\\
& -2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{hl}(w')\,U^{ma}(z)\,U^{mb}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
& -2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{hl}(x')\,U^{ma}(z)\,U^{mb}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
& +\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hb}(w)\,U^{hl}(x')\,U^{ma}(x)\,U^{md}(z)\,\rho^{l}(x')\,\rho^{e}(x)\\
& -2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
& +2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
& -\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{cde}\,U^{hb}(x')\,U^{hl}(w)\,U^{ma}(z)\,U^{md}(x')\,\rho^{l}(x)\,\rho^{e}(x')\\
& +\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{cde}\,U^{hb}(x')\,U^{hl}(x)\,U^{ma}(z)\,U^{md}(x')\,\rho^{l}(x)\,\rho^{e}(x')\\
& +\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{hl}(w)\,U^{ma}(z)\,U^{mb}(x')\,\rho^{l}(x)\,\rho^{e}(x')\\
& -\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{hl}(x)\,U^{ma}(z)\,U^{mb}(x')\,\rho^{l}(x)\,\rho^{e}(x')\\
& -\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^{b}(x)\,\rho^{e}(x')\\
& +\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,\rho^{b}(x')\,\rho^{e}(x)\\
& +\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^{b}(x)\,\rho^{e}(x')\\
& +\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
& -\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
& +\tfrac{3}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
& -\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
& +\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
& -\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
& +2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{ade}\,U^{hb}(x)\,U^{hd}(x')\,\rho^{e}(x')\,\rho^{c}(x)\Big]
\end{aligned} \qquad (5.14)$$

kernel structure	class	independent colour structures	differences
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$	charge–charge	12	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$	charge–charge	10	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	charge–charge	13	0
$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	charge–charge	14	0
$\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$	BFKL only	14	0
$\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x,z)$	BFKL only	10	0
$\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$	BFKL only	13	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x',x',z)$	BFKL only	10	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$	BFKL only	13	0

The sum. Collected, the sum is BFKL⊗LO, Eq. (3.2) before the charge average (orange / purple: 23 / 12):

$$
\begin{aligned}
\left.\frac{d^3N}{d^3k}\right|_{\text{BFKL}\otimes\text{LO}} = \mathcal{N}\log\vee\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,\rho^b(x')\,\rho^e(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(x)\,U^{hd}(x')\,\rho^e(x')\,\rho^c(x)\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x)\,\rho^e(x')\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\\
&+2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,\rho^a(x)\,\rho^a(x')\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w)\,U^{eh}(z)\,\rho^d(x)\,\rho^c(x')\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w)\,U^{eh}(z)\,\rho^d(x')\,\rho^c(x)\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{eh}(z)\,\rho^c(x)\,\rho^d(x')\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{eh}(z)\,\rho^c(x')\,\rho^d(x)\\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ca}(z)\,\rho^b(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ca}(z)\,\rho^b(x')\,\rho^c(x)\\
&-2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^e(x')\,\rho^c(x)\\
&-2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')\\
&+2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(w')\,U^{eh}(z)\,\rho^d(x)\,\rho^c(x')\\
&-2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(x')\,U^{eh}(z)\,\rho^d(x)\,\rho^c(x')\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^c(x)\,\rho^b(x')\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^b(x)\,\rho^c(x')\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ca}(z)\,\rho^b(x)\,\rho^c(x')\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ca}(z)\,\rho^b(x)\,\rho^c(x')\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{bde}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{bde}\,U^{hd}(x)\,U^{ha}(w')\,\rho^c(x')\,\rho^e(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dae}\,U^{hb}(w)\,U^{eh}(z)\,\rho^d(x')\,\rho^c(x)
\end{aligned}
$$

$$+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dae}\,U^{hb}(x)\,U^{eh}(z)\,\rho^{d}(x')\,\rho^{c}(x)\Big] \tag{5.15}$$

6 DGLAP

6.1 The poles in the $\log\frac{\vee}{\Lambda}$ terms cancel

Three genuine $\log\frac{\vee}{\Lambda}$ terms carry a pole: Group I Row II, Group I Row V 2b UV and Group II Row IV. This is the soft end ($\xi\to 1$) of the collinear divergence. Write

- $E(r)=\frac{2}{\epsilon}+2\gamma_E+\log\frac{r^2\mu^2_{\overline{MS}}}{4}$;
- $\mathcal{S}=e^{-ik\cdot(w'-w)}\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,[U(x')-U(w')]^{ab'}\rho^{b'}(x')\,[U(w)-U(x)]^{ab}\rho^{b}(x)$.

In Row V 2b rename $y\to x$, $z\to w$; in Group II Row IV rename $y\to w$, $y'\to w'$. Then, with the c.c. of Group I:

- Group I Row II: $+\pi N_c\big[E(x-w)+E(x'-w')\big]\mathcal{S}$;
- Group I Row V 2b UV: $-2\pi N_c\big[E(x-w)+E(x'-w')\big]\mathcal{S}$;
- Group II Row IV: $+2\pi N_c\,E(w-w')\,\mathcal{S}$.

All are in units of $\mathcal{N}\log\frac{\vee}{\Lambda}\int$. The sum is

$$\frac{1}{(2\pi)^3}\frac{g^4N_c}{16\pi^4}\frac{1}{k^+}\log\frac{\vee}{\Lambda}\int_{x,x',w,w'}\log\frac{(w-w')^4}{(x-w)^2(x'-w')^2}\,\mathcal{S}. \tag{6.1}$$

- The pole cancels: $+2-4+2=0$ in units of $\pi N_c\frac{2}{\epsilon}\mathcal{S}$. So do γ_E and μ.
- The remainder is part of JIMWLK⊗LO: it is the z integral of the terms whose colour does not depend on $U(z)$, done in the same scheme.
- So no DGLAP (collinear) logarithm multiplies $\log\frac{\vee}{\Lambda}$.

6.2 DGLAP from Group II, Row IV

At general $\xi=\frac{k^+}{p^++k^+}$ the pole of Group II Row IV is the real part of P_{gg} (`RowIV_FF_userform.pdf`). Write

- $\mathrm{LO}(\xi)=\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2k^+}\int_{x,y,x',y'}e^{-ik\cdot(y'-y)/\xi}\,X\cdot X'\,\mathcal{O}$;
- $X=\mathcal{K}(x-y)$, $X'=\mathcal{K}(x'-y')$;
- $\mathcal{O}=[U(y')-U(x')]^{e'c}\rho^{e'}(x')[U(y)-U(x)]^{ec}\rho^{e}(x)$.

Then:

$$\left.\frac{d^3N_{\mathrm{IV}}}{d^3k}\right|_{\mathrm{pole}}=\frac{\alpha_s}{2\pi}\Big(-\frac{2}{\epsilon}\Big)\bigg\{\int_{\frac{k^+}{\vee}}^1\frac{d\xi}{\xi^2}\,P^{(r)}_{gg,+}(\xi)\,\mathrm{LO}(\xi)+2N_c\Big[\log\frac{\vee}{\Lambda}-\log\frac{\vee}{k^+}\Big]\mathrm{LO}(1)\bigg\}, \tag{6.2}$$

$$P^{(r)}_{gg,+}(\xi)=2N_c\Big[\frac{\xi}{(1-\xi)_+}+\frac{1-\xi}{\xi}+\xi(1-\xi)\Big].$$

The $\delta(1-\xi)$ terms at the pole, in units of $\frac{\alpha_s}{2\pi}(-\frac{2}{\epsilon})N_c\,\mathrm{LO}(1)$:

source	coefficient
Group II Row IV	$2\log\frac{\vee}{\Lambda}-2\log\frac{\vee}{k^+}$
Group I Row II (with c.c.)	$-\frac{11}{3}+2\log\frac{\vee}{\Lambda}-2\log\frac{\vee}{k^+}-\log(1-\frac{k^+}{\vee})$
Group I Row V 2b UV (with c.c.)	$\frac{11}{3}-4\log\frac{\vee}{\Lambda}+4\log\frac{\vee}{k^+}$
charge renormalization of LO	$+\frac{11}{6}$
sum	$\frac{11}{6}-\log(1-\frac{k^+}{\vee})$

The $\log\frac{\vee}{\Lambda}$ $(2+2-4)$ and $\log\frac{\vee}{k^+}$ cancel; this is Eq. (6.1) again. The $\frac{11}{3}$ cancel between Group I Rows II and V (running coupling). What is left, $\frac{11N_c}{6}=\frac{\beta_0}{2}$, is the $\delta(1-\xi)$ part of P_{gg}. The remainder $-N_c\log(1-\frac{k^+}{\vee})$ is of order $k^+/\vee$ and is still open. Hence

$$\left.\frac{d^3N_g}{d^3k}\right|_{\mathrm{pole}}=\frac{\alpha_s}{2\pi}\Big(-\frac{2}{\epsilon}\Big)\int_{\frac{k^+}{\vee}}^1\frac{d\xi}{\xi^2}P_{gg}(\xi)\,\mathrm{LO}(\xi),\qquad P_{gg}(\xi)=2N_c\Big[\frac{\xi}{(1-\xi)_+}+\frac{1-\xi}{\xi}+\xi(1-\xi)\Big]+\frac{11N_c}{6}\delta(1-\xi). \tag{6.3}$$

The pole is absorbed into the fragmentation function, $D^{(0)}_{h/g}(\xi) = \int_\xi^1 \frac{dz}{z}\big[\delta(1-z) + \frac{\alpha_s}{2\pi}\frac{2}{\epsilon}P_{gg}(z)\big]D_{h/g}(\frac{\xi}{z},\mu)$. The μ independence of $\frac{d^3N_h}{d^3k}$ then gives

$$\boxed{\mu^2\frac{\partial}{\partial\mu^2}D_{h/g}(\xi,\mu) = \frac{\alpha_s(\mu)}{2\pi}\int_\xi^1 \frac{dz}{z}P_{gg}(z)\,D_{h/g}\Big(\frac{\xi}{z},\mu\Big)} \tag{6.4}$$

with the natural scale $\mu = c_0/|y'-y| \sim |\mathbf{k}|$, $c_0 = 2e^{-\gamma_E}$.

7 Three and four ρ's

- **Three ρ's.** The list "All three ρ term with c.c." (152 terms, classes α, α', β, β') is identical to the list of `ThreeRho_cancellation_v3.pdf`.
  - It is checked again here with random SU(3) Wilson lines ($U^{bc} = U^{cb}$) and charges, after full symmetrization.
  - There are 0 non-cancelling kernel monomials; the largest residual is 2×10^{-15}, against single terms of $O(1)$.
  - Its two-ρ remainders are your Sec. 2, which is checked in Sec. 8.
- **Four ρ's.** The four-ρ terms of `4rho.tex` cancel after symmetrization: 4.7×10^{-15} (scale 66). Their three-ρ remainders are d^3N_a, d^3N_b (10^{-14}). Their two-ρ remainders are the `4rho.tex` lists of Eq. (4.9) and your "From 4ρ's".
- **Display issues (M3, M4)**, which do not enter the lists:
  - `Group_I.tex` Eq. (1.5);
  - the top of `Evolution_three_rho.tex`, Rows II and III.

  See the summary.

8 Checks

Scripts in `final_check/`:

check	result
Sec. 2 of `Evolution_terms.tex`, row by row, vs the checked lists (`compare_sec2.py`): differing kernel monomials	A,B: 0, 3:I.II: 0, 3:I.III: 0, D: 0, 3:I.IV: 0, 3:I.V: 0, 3:II.I: 0, 3:III.I: 0, 3:III.II: 0, 3:III.III: 0, 3:III.V: 0, 3:III.VI: 0
the same, pointwise (largest relative difference)	6×10^{-15}
genuine terms of Sec. 1 vs the corrected genuine terms (`genuine_new.py`)	$\le 10^{-15}$
Row V 2b: UV term + counterterm, pointwise	0 (10^{-18})
all two-ρ $\log\frac{\vee}{\Lambda}$ terms of `Evolution_terms.tex` vs JIMWLK⊗LO (`total_check.py`)	rel. 1.7×10^{-1} (6 of 26 monomials differ)
the same with the `4rho.tex` lists	rel. 6×10^{-16}
exact, by independent colour structure (`render.py`)	0 of 154 differ
$\log\vee$ terms vs BFKL⊗LO, exact (`render.py`)	0 of 109 differ
JIMWLK formula as sent / corrected vs JIMWLK⊗LO (`targets_check.py`)	0.22–0.47 / 2×10^{-15}
BFKL formula (App. E) vs BFKL⊗LO	8×10^{-15}
three-ρ list: identical to v3; cancellation (`three_cancel.py`)	yes; residual 2×10^{-15}